

effect in h periods since the treatment). We define the two estimation targets mathematically in this section.

Let $\Omega = \{it\}$ denote the set of all observations from repository i and month t , Ω_1 denote the set of treated observations, and Ω_0 denote the set of untreated (i.e., never-treated and not-yet-treated) observations. Let Y_{it} denote the *actual* outcome of interest for repository i at month t , and $Y_{it}(0)$ denote the *potential* outcome for repository i at month t if it is never treated. ATT is defined as the average of this causal treatment effect on all treated observations:

$$ATT = \frac{1}{|\Omega_1|} \sum_{it \in \Omega_1} \mathbb{E}[Y_{it} - Y_{it}(0)] \quad (2)$$

Let $\Omega_{1,h}$ denote the set of all treated observations h time periods after the treatment; ATT_h is defined as the average of the causal treatment effect in that specific horizon h :

$$ATT_h = \frac{1}{|\Omega_{1,h}|} \sum_{it \in \Omega_{1,h}} \mathbb{E}[Y_{it} - Y_{it}(0)] \quad (3)$$

3.3.3 The Borusyak et al. [29] Estimator. There are many possible methods to estimate ATT and ATT_h defined in Section 3.3.2, among which the two-way fixed effects (TWFE) estimator is most commonly used in early econometric studies. However, recent research shows that TWFE may produce biased estimates in the staggered adoption setting if treatment effects are heterogeneous over time [23, 44, 54]. To address this known limitation, we use the Borusyak et al. [29] imputation estimator, a state-of-the-art estimator designed explicitly for robust and efficient estimation in the staggered adoption setting, with this two-step process:

Step 1: Impute counterfactual outcomes. The estimator fits a counterfactual outcome regression model, using only untreated observations Ω_0 (i.e., pre-adoption observations for treated repositories and all observations for never-treated controls):

$$\hat{Y}_{it}(0) = \hat{\mu}_i + \hat{\lambda}_t + \hat{\Gamma}'Z_{it} + \epsilon_{it} \quad (4)$$

where $\hat{\mu}_i$, $\hat{\lambda}_t$ represent per-repository and per-month fixed effects; Z_{it} includes time-varying covariates (Section 3.2.2), and $\epsilon_{it} = Y_{it} - \hat{Y}_{it}$, $it \in \Omega_0$ is the error term when fitting this model on Ω_0 . The use of only Ω_0 in this step ensures that counterfactual predictions are not contaminated by treatment effects as in the TWFE estimator. It is also important to note that all repository-invariant confounders (e.g., team culture, domain, language) are effectively controlled by per-repository fixed terms $\hat{\mu}_i$ and all time-invariant confounders (e.g., industry trends, platform changes, and seasonal patterns) are effectively controlled by per-month fixed terms $\hat{\lambda}_t$ in this step.

Step 2: Compare actual to counterfactual. For each repository-month post-adoption ($it \in \Omega_1$), the estimator predicts the potential outcome from the counterfactual outcome model: $\hat{Y}_{it}(0) = \hat{\mu}_i + \hat{\lambda}_t + \hat{\Gamma}'X_{it}$. We replace the $Y_{it}(0)$ in Equations 2 and 3 with the estimated $\hat{Y}_{it}(0)$, to get the final ATT and ATT_h estimations.

Finally, to assess the plausibility of the parallel trend assumption, Borusyak et al. [29] advises fitting an alternative model of Y_{it} for untreated observations Ω_0 with additional observables. In our paper, we follow the most typical convention and fit the following model with additional dummies before the onset of treatment:

$$Y_{it} = \hat{\mu}_i + \hat{\lambda}_t + \hat{\Gamma}'Z_{it} + \sum_{h=-k}^{-2} \hat{\tau}_h \mathbf{1}[t = E_i + h] + \epsilon_{it} \quad (5)$$

where E_i means the time of treatment for repository i and $\mathbf{1}[t = E_i + h]$ is equal to 1 if and only if the current time t is h months away from treatment (0 otherwise). Then, we use heteroscedasticity- and cluster-robust Wald tests [29] to test the joint null hypothesis that $\hat{\tau}_h = 0$ for $h = -k, \dots, -2$. We drop $h = -1$ due to potential anticipation concerns (the developer may try use Cursor before adding Cursor rule files in the immediate month before). One way of viewing the above pre-trend testing procedure is a placebo test, in which τ_h estimates ATT_h for $h < 0$, which should be zero if the treatment has not yet happened. The estimated $\hat{\tau}_h$ for $h < 0$ and ATT_h for $h > 0$ are often combined into *event study plots* (e.g., Figure 3), in which the dynamic effect of treatment is visualized.

Note that the Borusyak et al. [29] imputation estimator has many alternative specifications with varying assumptions, and we merely describe the version we used in our paper. We refer interested readers to the original paper [29] regarding alternative specifications and the mathematical assumptions behind them. The Borusyak et al. [29] estimator is also not the only option, and we present results from alternative DiD estimators (TWFE and Callaway and Sant'Anna [31]) in the Appendix as additional robustness checks.

3.4 Testing Velocity & Quality Interactions

While DiD estimates treatment effects on individual outcomes, it does not capture temporal dynamics *between* outcomes. However, it is known that velocity and quality outcomes interact in our setting [28, 38]. Plus, our DiD results (Section 4) also suggest interactions, showing that the adoption of Cursor leads to non-sustained velocity increases and sustained quality declines: Development velocity increases may cause rapid technical debt accumulation, which may subsequently decrease velocity.

To test for dynamic relationships and bidirectional causality, we use the *generalized method of moments* (GMM) [58] to obtain consistent estimates when variables are potentially endogenous (correlated with unobserved errors). The key insight is to use *instrumental variables*—correlated with endogenous regressors but uncorrelated with errors—to identify causal effects. In panel data, lagged values serve as natural instruments, assuming past values influence current values but are uncorrelated with current shocks [22].

In our study, we use Arellano-Bond dynamic panel GMM [22], suited for: (1) dynamic dependence (current outcomes depend on past); (2) potential bidirectional causality; (3) short time series with many entities. To test a causality direction $X_t \rightarrow Y_t$ while accounting for Cursor adoption D , we estimate the following regression:

$$Y_{it} = \hat{\mu}_i + \hat{\lambda}_t + \hat{\rho}Y_{i,t-1} + \hat{\beta}D_{it} + \hat{\gamma}X_{it} + \hat{\Gamma}'Z_{it} + \epsilon_{it} \quad (6)$$

where $Y_{i,t-1}$ captures outcome persistence; D_{it} is a dummy representing Cursor adoption; X_{it} represents the potentially endogenous regressor of interest; the remaining follows Equation 4. During estimation, historical values of X_{it} (e.g., a linear combination of $X_{i,t-2}$ and $X_{i,t-3}$) are used as instrumental variables.

Specifically, we test the following temporal interactions:

$$\begin{aligned} \text{Lines Added}_{it} &\rightarrow \text{Static Analysis Warnings}_{it} \\ \text{Lines Added}_{it} &\rightarrow \text{Code Complexity}_{it} \\ \text{Static Analysis Warnings}_{it} &\rightarrow \text{Lines Added}_{i,t+1} \\ \text{Code Complexity}_{it} &\rightarrow \text{Lines Added}_{i,t+1} \end{aligned} \quad (7)$$